Board B — Floating gate driver

Single Adafruit Perma-Proto 1/2 carrier for the 6N137 / 4N35 optocouplers, Q2 fail-off, TC4426 gate driver, and Q3 K1 coil switch. Earth-side LED inputs occupy columns 1–5; the floating gate-driver region occupies columns 6–30. Floating power and inter-board links leave through screw terminals on the board edges.

> **Draft.** View with U1 (6N137) on the left and U4 (4N35) on the right, matching the as-built photo. Vertical stacks (**T1–T3** upper left, **T6–T7** lower left) are numbered top → bottom. Horizontal screws (**T4–T5** top edge) are numbered left → right. **T1–T11** are as-built on the populated unit. Float rails: top **+** is `+12V_GD`, bottom **+** is `+5V_ISO`, both **−** rails share `−70V`. On-board jumper paths and pin-by-pin IC wiring stay out of scope here; see Motor_Driver_Wiring.md.

> **Two galvanic domains.** Columns 1–5 are the earth LED island: `PWM`, `DIR`, and `MCU_GND` from Board A (T-Display-S3 GPIO16 / GPIO21 and earth return). Those nets are earth-referenced only. Columns 6–30 are floating, referenced to `−70V`. U1 and U4 translate the command signals optically; no copper joins the domains. Never tie `MCU_GND` to `−70V`. After HDR-15-5 #2 and HDR-15-12 negatives bond to `−70V`, treat both floating supplies as hazardous relative to earth. Do not clip an earth-referenced scope ground to `−70V`, the gate node, or `SW`.

1. Placement

View the board from above with U1 on the left and U4 on the right, as in the as-built photo. U3 (TC4426) sits in the center gutter; Q2 is beside U3; Q3 is beside U4. The isolation barrier falls at approximately column 5: a physical break in the proto buses separates the earth LED island from the floating region. The upper-left corner holds a vertical 3-position block (**T1–T3**). The earth-side Board A cable lands on **T4** (`PWM`) and **T5** (`MCU_GND`). The lower-left stack (**T6–T7**) carries HDR-15-5 #2. The upper-right block (**T8–T9**) takes `DIR` from Board A and drives PWM out to Board C. The lower-right block (**T10–T11**) wires the Phoenix K1 coil.

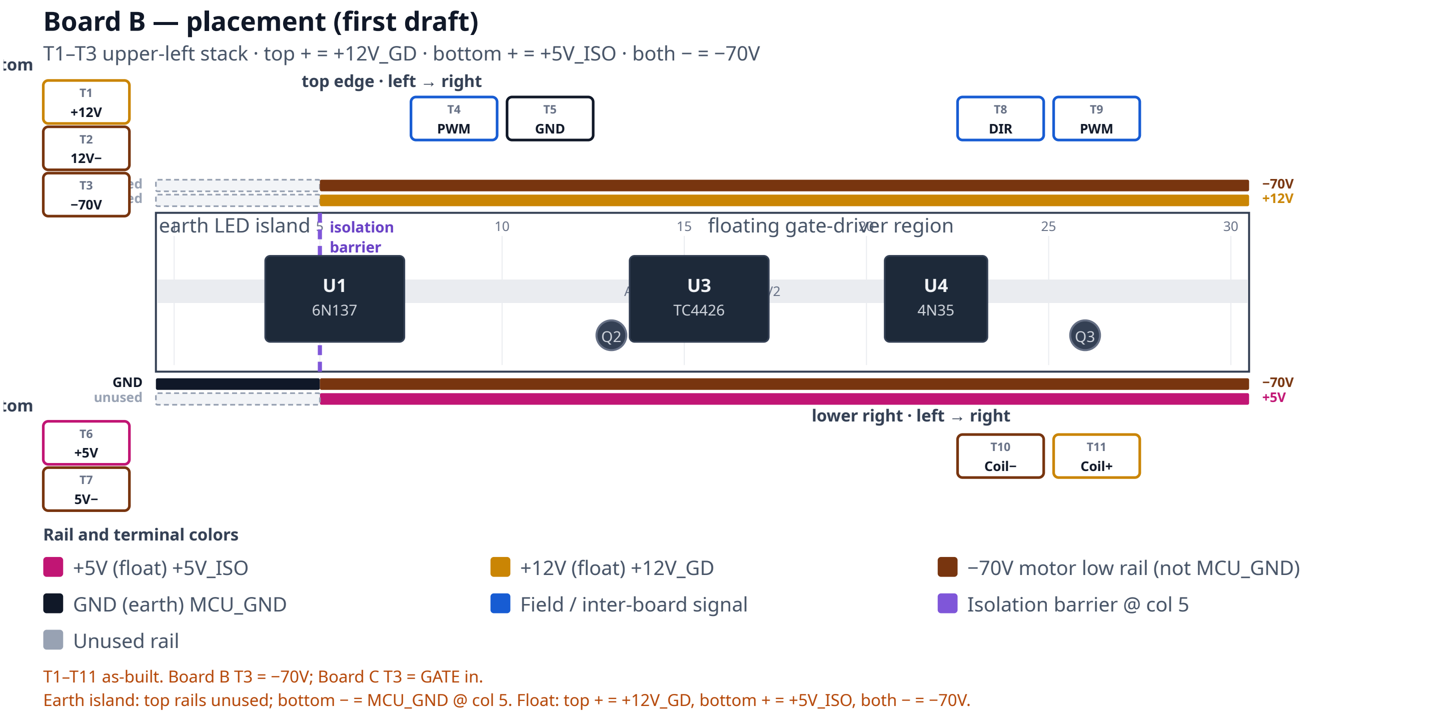


Figure 1. Grid-true placement draft. Left-edge stacks (**T1–T3**, **T6–T7**) sit beside the Perma-Proto outline, matching the physical board. Purple dashed line at column 5 is the isolation barrier. LED resistors, Q2 bias, TC4426 decoupling, and the 100 Ω gate resistor are on-board only.

2. Rails

Each Perma-Proto 1/2 has four full-width buses (top −, top +, bottom −, bottom +). A single bus cannot carry two domains; the isolation barrier at column 5 keeps earth-side copper out of the floating region. On the as-built unit, the bottom **−** rail on the earth island (column 5) lands `MCU_GND`. In the floating region, both **−** rails are `-70V`, the top **+** rail is `+12V_GD`, and the bottom **+** rail is `+5V_ISO`.

Rail (U1 left, U4 right)	Earth island (cols 1–5)	Floating region (cols 6–30)	Diagram color
Top −	unused	`-70V` (as-built)	■ −70V
Top +	unused	`+12V_GD` (as-built)	■ +12V_GD
Bottom −	`MCU_GND` (as-built @ col 5)	`-70V` (as-built)	■ −70V / ■ GND
Bottom +	unused	`+5V_ISO` (as-built)	■ +5V_ISO

3. What attaches

Power

Board B draws from two floating DIN supplies whose negatives are bonded only to `-70V`. The command cable from Board A carries no floating power — only earth-referenced GPIO and `MCU_GND`.

Land Mean Well **HDR-15-12** on the upper-left stack: positive on **T1** (`+12V_GD`) and negative on **T2** (12 V supply return — bonded to `−70V`, not `MCU_GND`). That feed drives the top **+** rail (U3 VDD, Q2 bias, K1 coil branch).

Land Mean Well **HDR-15-5 #2** on the lower-left stack: positive on **T6** (`+5V_ISO`, top screw) and negative on **T7** (bottom screw — floating return bonded to `−70V`, not `MCU_GND`) → bottom **+** rail → U1 VCC and VE. Do not connect either floating supply negative to `MCU_GND` or protective earth.

T3 (Board B) is the `−70V` landing for the short harness to Board C (Q1 source / U3 GND return). Do not confuse it with Board C **T3**, which receives PWM from Board B **T9**.

Screw terminals

Numbering follows Figure 1. Vertical stacks are top → bottom; horizontal screws are left → right on that edge.

Upper left — T1–T3 (top → bottom, as-built)

Terminal	Label	Net	Attaches
T1	**+12V**	`+12V_GD`	**Power in.** Mean Well HDR-15-12 (+) → top **+** rail
T2	**12V−**	`−70V`	**Power in.** Mean Well HDR-15-12 (−). Floating supply return bonded to `−70V`. This is not `MCU_GND`.
T3	**−70V**	`−70V`	Board B → Board C shared motor low rail (Q1 source / U3 pin 3). Keep this link short.

Top edge — T4–T5 (left → right, as-built)

Terminal	Label	Net	Attaches
T4	**PWM**	`PWM` (GPIO16)	Board A **T5** ← T-Display-S3 GPIO16. Earth island only → U1 LED through 220 Ω on-board. Optically isolated to the floating domain.
T5	**GND**	`MCU_GND`	Board A **T12** ← T-Display earth return. U1 / U4 LED cathodes. Never tie to `−70V` or floating supplies.

Lower left — T6–T7 (top → bottom, as-built)

Terminal	Label	Net	Attaches
T6	**+5V**	`+5V_ISO`	**Power in.** Mean Well HDR-15-5 #2 (+) → bottom **+** rail → U1 VCC / VE
T7	**5V−**	`−70V`	**Power in.** Mean Well HDR-15-5 #2 (−). Floating supply return bonded to `−70V`. Not `MCU_GND`.

Upper right — T8–T9 (left → right, as-built)

Terminal	Label	Net	Attaches
T8	**DIR**	`DIR` (GPIO21)	Board A **T6** ← T-Display-S3 GPIO21. Earth island → U4 LED (270 Ω on-board). LED return shares **T5** `MCU_GND` — no separate earth screw for DIR. Optically isolated to Q3 / K1 coil.
T9	**PWM out**	`PWM` / gate	TC4426 OUT B through 100 Ω (on-board) → Board C **T3**. Floating domain. Return via Board B **T3** `−70V`. Keep harness short.

Lower right — T10–T11 (left → right, as-built)

Terminal	Label	Net	Attaches
T10	**Coil−**	K1 coil (−) / Q3 drain	Phoenix **PLC-RSC-12DC/21-21** coil (−) via field wire. Low side switched by Q3 to `−70V` on-board.
T11	**Coil+**	`+12V_GD`	Phoenix K1 coil (+) from `+12V_GD` rail (HDR-15-12 / top **+** bus). D5 flyback (1N4007) is installed across these terminals on the as-built unit; cathode/band toward **T11**.

K1 coil (on-board driver, DIN relay)

Q3 switches the Phoenix K1 coil low-side on the floating domain. Field wires to the DIN relay coil land on **T10** (Coil−) and **T11** (Coil+). Do not route coil current through the Board B → Board C PWM link (**T9**).

Signal	Net	Attaches
K1 coil +	`+12V_GD`	**T11** → Phoenix PLC-RSC-12DC/21-21 coil (+)
K1 coil −	Q3 drain	**T10** ← relay coil (−); Q3 source → `−70V`
PWM command	`PWM`	T-Display-S3 GPIO16 → Board A **T5** → Board B **T4** → U1 LED (220 Ω on-board); optically isolated to Q2 / U3 / gate
Direction command	`DIR`	T-Display-S3 GPIO21 → Board A **T6** → Board B **T8** → U4 LED; return via **T5** `MCU_GND`

Coil and contact detail: K1 floating coil option.

On-board only (not on screw terminals)

These subsystems stay on the Perma-Proto and are wired per Motor_Driver_Wiring.md:

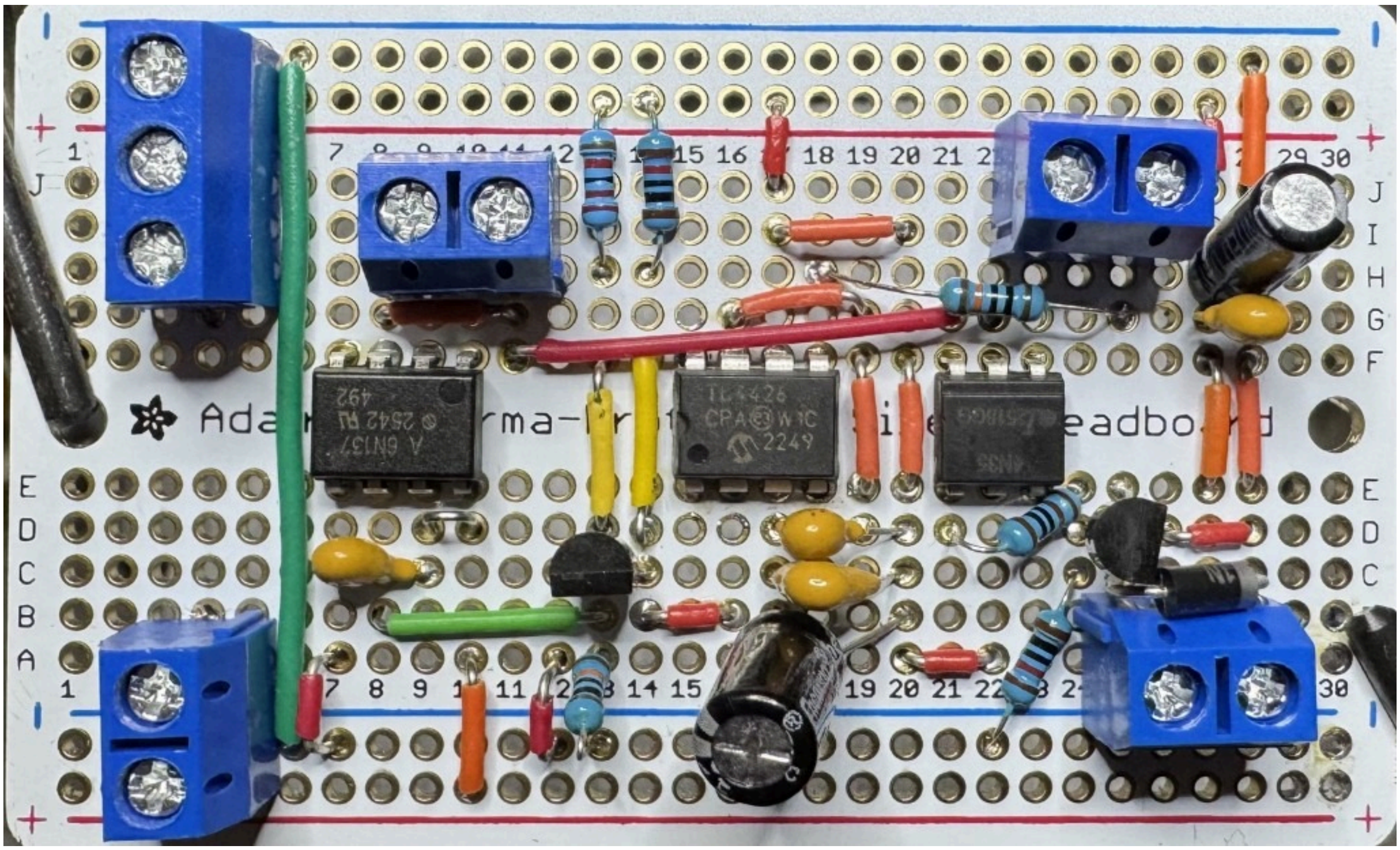
- U1 6N137 — PWM isolation; pin 6 → Q2 base
- Q2 PN2222A — fail-off inverter (Rb2, Rbe2, Rc2 on-board)
- U3 TC4426 — dual inverting gate driver; OUT B → 100 Ω → Board C gate path
- U4 4N35 + Q3 2N7000TA — direction opto and coil low-side switch
- 220 Ω (U1 LED) and 270 Ω (U4 LED) beside the optocouplers
- U3 decoupling: 100 nF, 1 µF across pins 6–3; 47 µF bulk nearby

Inter-board cables

Cable	Board B landing	Domain
Board A → Board B	**T4** (`PWM` in), **T5** (`MCU_GND`), **T8** (`DIR`)	Earth only; translated optically by U1 / U4
Board B → Board C	**T3** (`−70V`) + **T9** (PWM out → Board C **T3**)	Floating; keep short
Board B → DIN K1 coil	**T10** (Coil−), **T11** (Coil+ / `+12V_GD`)	Floating

4. As-built photo

Line art is the labeled figure. The photo shows the unit that was built.



Populated carrier. U1 (6N137) on the left, U3 (TC4426) in the center, U4 (4N35) on the right. Isolation barrier at column 5.

Related

This page is a chapter of the Wiring Guide. Electrical wiring truth for the motor stack remains the gate-driver schematic, not the placement drawing.